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In-vivo observing effects of the microgravity environment on zebrafish eyes using optical coherence tomography

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ABSTRACT

Microgravity, vacuum, and high-intensity ultraviolet waves are widely known characteristics of space. These different environments from the earth affect physical changes including ocular tissue changes while astronauts stay in the universe. The changes in ocular tissue in the space environment, also known as visual impairment intracranial pressure (VIIP) syndrome, including fundus optic disc edema, hyperopic drift, choroidal folds, cotton spots, and permanent fundus damage could influence astronauts' vision system and ability of space operations. Especially, hyperopic drift by posterior flattening and folded retina by choroidal folds are reported to affect the retina's structures as a vision sensor directly. To investigate microgravity's effect on ocular tissues and vision, previous research on earth are used special facilities and various microgravity simulators, including head-down tilt bed-rest and random positioning machines. This study suggests that an experiment expose wild-type zebrafish to microgravity using a rotary cell culture system (RCCs) applied to experiments using cell and zebrafish's embryos in microgravity. Unlike previous research using zebrafish's embryos and larva, adult and growing zebrafish were employed in this study for observing ocular changes in simulated space environments. After exposing zebrafish to microgravity, in-vivo zebrafish's eye images were acquired by custom-built optical coherence tomography (OCT). This research for presenting the new method for small animal experiments in microgravity environments could be applied to investigate the influence of staying in the universe on an animal model with ophthalmic diseases.

Keywords: Zebrafish, Microgravity, visual impairment intracranial pressure, optical coherence tomography, OCT

1. INTRODUCTION

The universe's environment is one of the obstacles to exploring and traveling outside of the earth. High technology-based spacesuits and specialized apparatus make astronauts overcome some obstacles including high-intensity ultraviolet and vacuum. However, microgravity still remains an unsolved issue affecting astronauts' health. According to NASA's research report and earlier study based on comparisons of medical examination between before and after space exploring, some symptoms are reported in the astronauts' musculoskeletal and ophthalmic systems[1-3]. Especially, damaged vision system, called visual impairment intracranial pressure (VIIP), is a critical and chronic aftereffect involving persistent changes with different degrees of severity in post-flight. VIIP symptoms are optic disc edema, globe flattening, choroidal folds, hyperopic shifts, and intracranial pressure increase. These aftereffects are not lead to blindness in the short term, but the prognosis of the patients with longitudinal and chronic progress could become severe[1, 4]. Although the exact pathogenesis of VIIP are not demonstrated, previous researches have presumed that fluid changes affected by microgravity lead to abnormally increased intracranial and intraocular pressure[4]. Abnormally increased intraocular pressure (IOP) is one of the most critical risk factors for ophthalmic diseases including glaucoma[5-7]. The risk of the high-IOP is not confining to the adult, so all age humans should care. The research about the aftereffect of exposure by universe environment is only restricted to adults because all astronauts are adults in the previous research so far. To prepare for the future that people of all ages and both sexes could stay and travel in the universe, we suggest that all age animal eyes exposed by microgravity should be observed by in-vivo monitoring. Although the previous studies investigated

microgravity effects on zebrafish embryos through comparison between control and microgravity affected groups after their growth, there is no research using growing or adult zebrafish to study microgravity effects[8-10]. For these reasons, we suggest this study inflict microgravity on different ages zebrafish and observe them through in-vivo imaging.

2. METHODS AND MATERIALS

2.1 Apparatus for exposing microgravity to zebrafish

The Rotary Cell Culture system (RCCs) is one of the equipment that can inflict microgravity on cell and embryo level organisms[11, 12]. Comparing the other microgravity simulators, such as head-down tilt bed-rest and random positioning machines, the inner side of the RCCs chamber is filled with water. Due to this characteristic, RCCs has been used in previous research exposing microgravity to zebrafish embryos and is chosen in this study to use growing and adult zebrafish[8, 11, 12]. The principle of the RCCs is based on the force combination during system operation[13, 14]. When the vessel stops rotating, the gravitational vector force mainly affects the organism in the vessel. While the system is working, additional forces, including the viscosity of water as well as the centripetal and centrifugal force by rotation, are created and diminish the influence of the gravitational vector. It is essential for mimicking a microgravity environment that the combined power converge towards zero through the balance of these forces. If specific energy is biased, simulating microgravity is failed. For instance, higher centrifugal force than other powers leads to the status shown in Fig. 1(c). According to the previous study, rotating speed control can adjust the force equilibrium, including centripetal and centrifugal force.

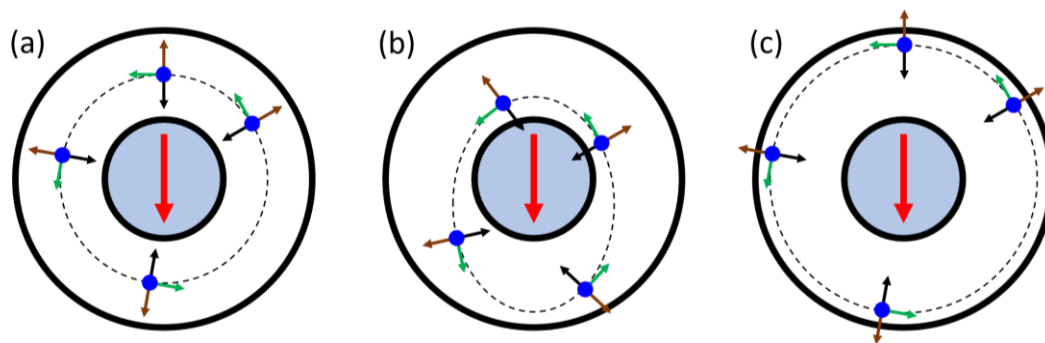


Figure 1. Cross-sections of the orbit of the experimental model in the vessel by rotating speed. (a): microgravity mimicked via specific rotating speed, (b): The case that gravity is overwhelming in force combination by too slow rotating speed, (c): The case that centrifugal force is overwhelming in force combination by too fast rotating speed. The blue circle shows the position of the experimental target. The red arrow shows the direction of gravity. The green arrow means the power of instant direction of rotating. Brown arrow is a centrifugal force. The black arrow is a centripetal force. Sky blue circle is a Teflon central core of the vessel.

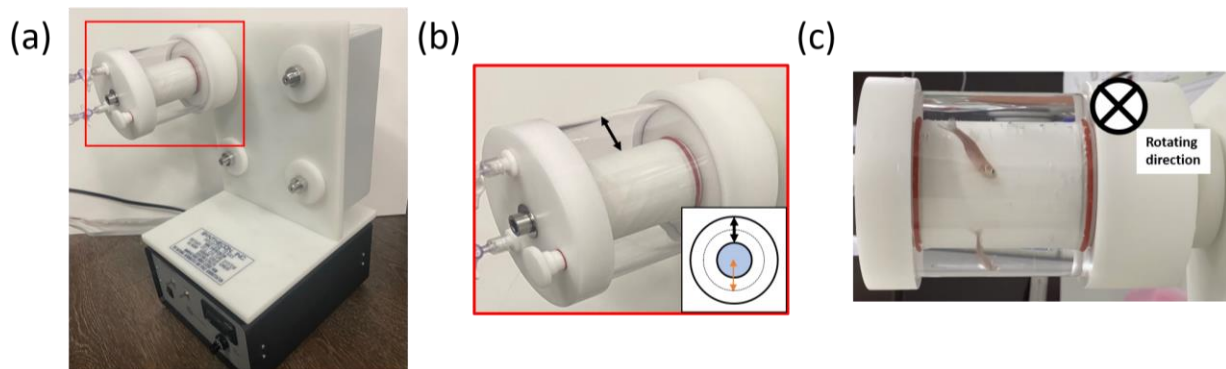


Figure 2. RCCs system explain. (a): whole system, (b): RCCs diameter explain (black arrow: 2 cm, orange arrow: 3.75 cm), (c): capture of actual operating video.

2.2 In-vivo imaging system

Most symptoms of VIIP occur in the posterior segment of the eye including the retina. Custom-built optical coherence tomography (OCT), a useful in-vivo retina imaging tool, is used to observe the aftereffect of microgravity exposure. This system was optimized at our previous study through opticstudio®[15]. The improvement compared with the former study is prism set for dispersion matching.

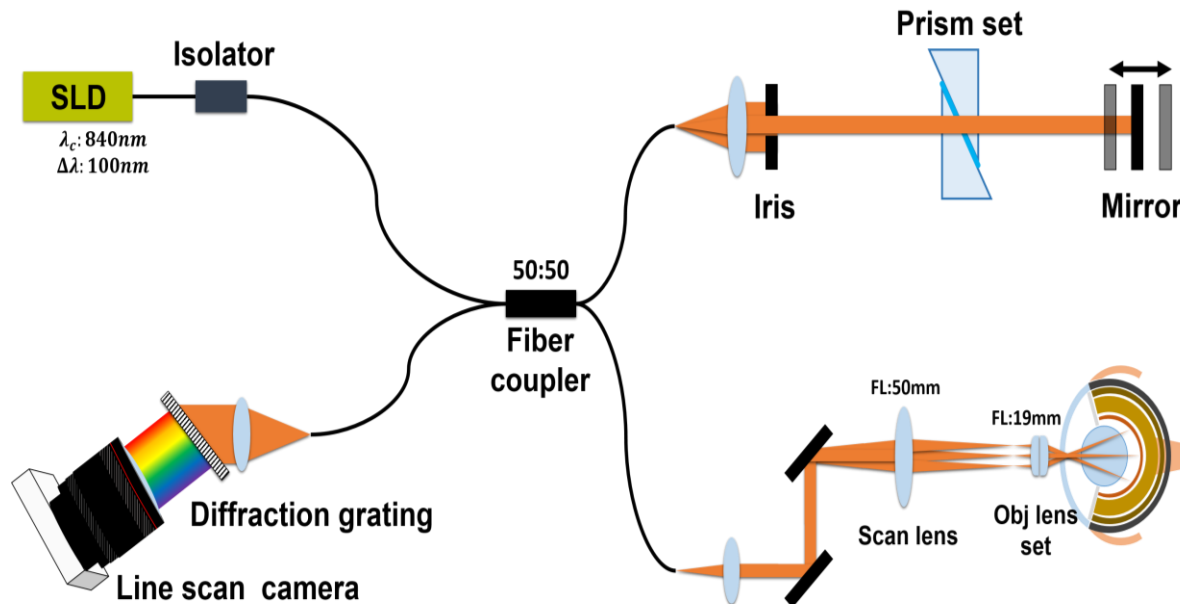


Figure 3. Schematic of the used OCT.

2.3 Experiment protocol

All zebrafish retinal imaging was performed in anesthesia status using the tricaine mesylate solution based on ZFIN (an online biological database of information about the zebrafish). The imaging cycle is five weeks, meaning before and after microgravity exposure. The rotating speed is 42.5 rpm referring to previous research investigating RCCs rotating speed. We used three different ages adult wild-type zebrafish (19 months, 9 months, and 3 months). The eldest wild-type zebrafish are purchased local aquarium and the other wild-type zebrafish was provided by the Zebrafish Center for Disease Modeling (ZCDM), Korea. All experiments have complied with animal protocol approval approved by the IACUC of Inha University (Approval number: 200901-717).

3. RESULTS

We confirmed the performance of our imaging system as an observing tool for zebrafish retina morphology changes through the possibility of retinal layer distinction. Retina images of zebrafish, taken before microgravity exposure, are shown in Fig. 4. Fig. 4(a), 4(b), 4(c) are single B-scan images of each B-scan set in the three different ages adult zebrafish. And Fig. 4(d), 4(e), 4(f) are post-processed images with averaging after motion correction. Motion correction and averaging processes are performed using Image J plugins (moco adjustment and z-projection). Comparing Fig. 4(e), 4(f) with Fig. 4(d), Fig. 4(d) is blurry due to the large motion by breathing of 19 months zebrafish during in-vivo imaging. However, it is not a problem to distinguish each retinal layer. The corresponding retinal layers are represented in Fig. 4(d).

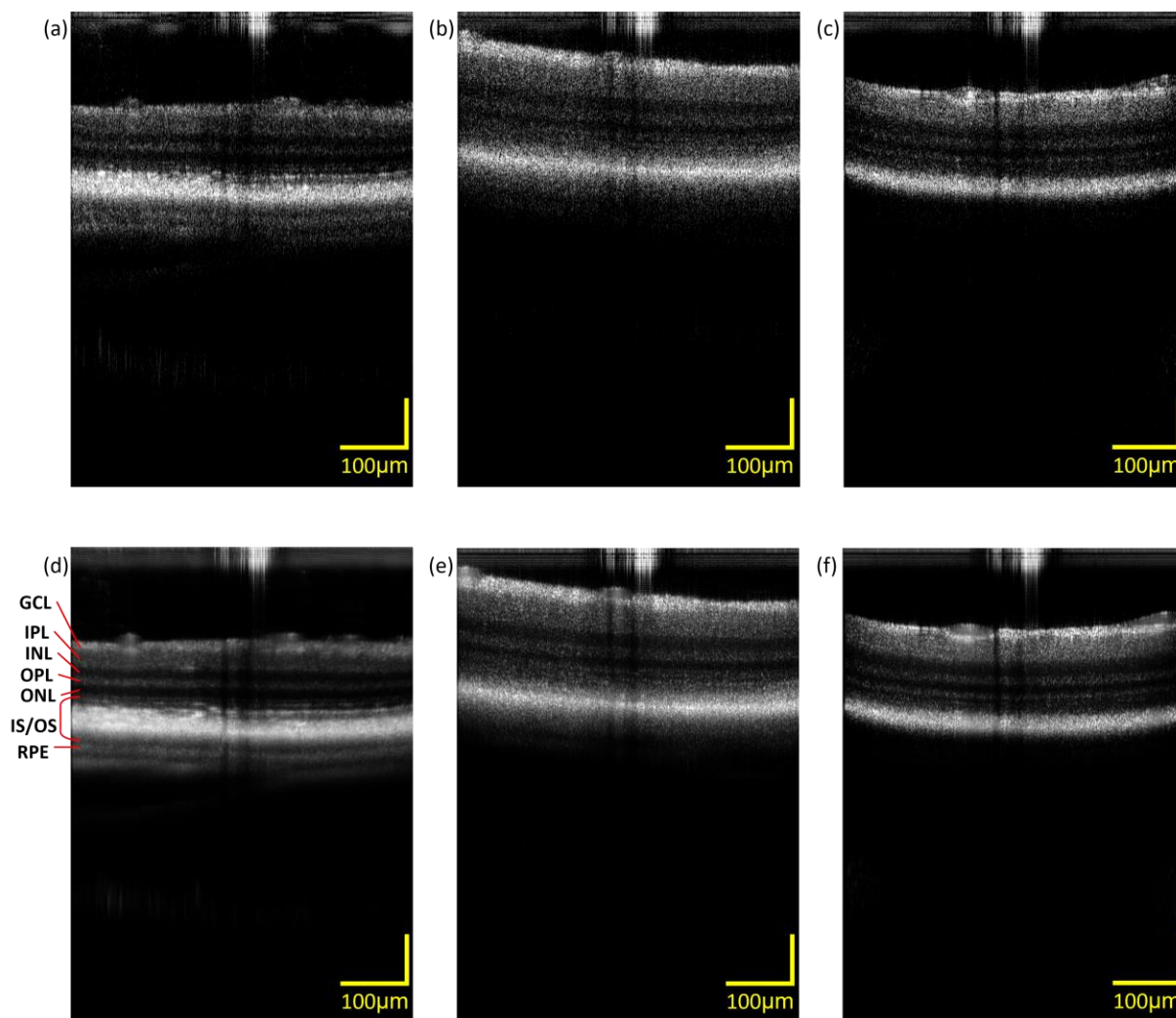


Figure 4. Cross-sectional images of zebrafish retina. (a): a single B-scan of 19 month zebrafish, (b): a single B-scan of 9 months one, (c): a single B-scan of 3months one, (d): post-processd image of B-scan set (a), (e): post-processd image of B-scan set (b), (f) post-processd image of B-scan set (c).

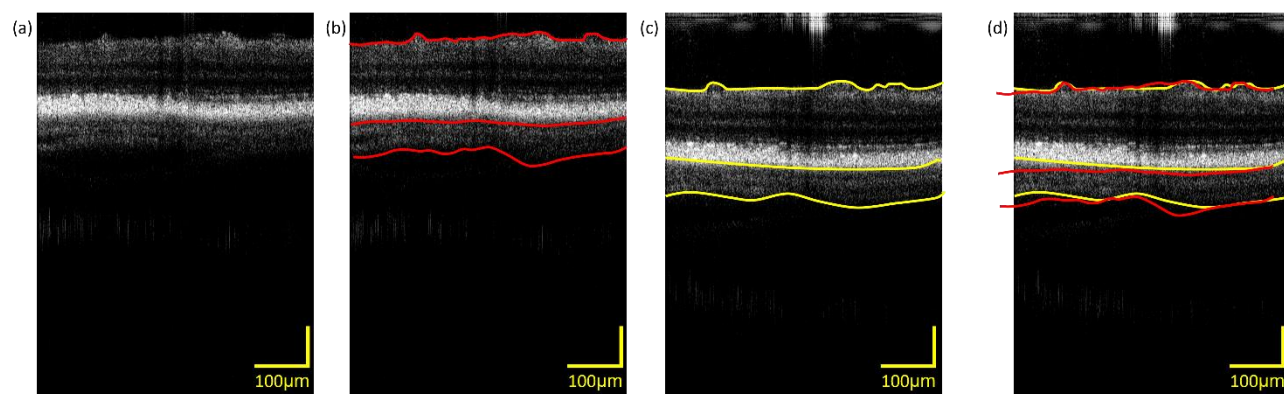


Figure 5. Comparison between before and after microgravity exposure. (a): a single B-scan of 19 months zebrafish, (b): manually segmented retinal layer line of (a)(red line), (c): manually segmented retinal layer line of Fig. 4(a)(yellow line), (d): red line overlap with (c).

After a lapse of five weeks, we acquired retinal images of 19 months zebrafish. Fig. 5(d) shows the comparison between before and after microgravity exposure using a manually segmented layer line. Even though we expected significant aftereffects including globe flattening and choroidal folds of VIIP symptoms, however, it is insignificant.

4. CONCLUSION

In this study, we inflict microgravity on zebrafish and observe their retina using optical coherence tomography. Through this experiment, we confirmed performance for the zebrafish retina imaging of our system and observed the aftereffect of microgravity exposure. Even though two different ages zebrafish (9 and 3 months) retired due to the death of an unknown cause, we found slight differences before and after microgravity exposure. However, the differences are not enough to analyze the influences of microgravity. We presumed that an insufficient exposure time is one of the reasons that could not create VIIP on the zebrafish. According to NASA's reports, VIIP onset times for individual astronauts are various. Based on this study, we will adjust the microgravity inflicting conditions, imaging cycles, and animal ages for making microgravity-affected zebrafish. If the protocol for inducing VIIP is established, It is possible that the research use zebrafish with various ophthalmic diseases in the simulated universe environment.

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REFERENCES

- [1] G. C. Alexander DJ, Hamilton DR, Lee SMC, Mader TH, Otto C, Oubre CM, Pass AF, Platts SH, Scott JM, Smith SM, Stenger MB, Westby CM, Zanello SB, [Evidence Report: Risk of Spaceflight-Induced Intracranial Hypertension and Vision Alterations.] NASA Lyndon B. Johnson Space Center, Houston, TX, USA(2012).
- [2] O. J. Juhl, E. G. Buettmann, M. A. Friedman *et al.*, "Update on the effects of microgravity on the musculoskeletal system," *npj Microgravity*, 7(1), 28 (2021).
- [3] L.-F. Zhang, and A. R. Hargens, "Spaceflight-Induced Intracranial Hypertension and Visual Impairment: Pathophysiology and Countermeasures," *Physiological Reviews*, 98(1), 59-87 (2018).
- [4] C. O. Jennifer A. Fogarty, Erik Kesrman, Cherie Oubre, Jimmy Wu [The Visual Impairment Intracranial Pressure Summit Report] NASA Johnson Space Center, Houston, TX, USA(2011).
- [5] I. Goldberg, "Relationship Between Intraocular Pressure and Preservation of Visual Field in Glaucoma," *Survey of Ophthalmology*, 48(2, Supplement), S3-S7 (2003).
- [6] M. B. Sultan, S. L. Mansberger, and P. P. Lee, "Understanding the Importance of IOP Variables in Glaucoma: A Systematic Review," *Survey of Ophthalmology*, 54(6), 643-662 (2009).
- [7] C. Leydolt, O. Findl, and W. Drexler, "Effects of change in intraocular pressure on axial eye length and lens position," *Eye*, 22(5), 657-661 (2008).
- [8] S. C. Edsall, and T. A. Franz-Odenaal, "An assessment of the long-term effects of simulated microgravity on cranial neural crest cells in zebrafish embryos with a focus on the adult skeleton," *PLoS One*, 9(2), e89296 (2014).
- [9] N. Shimada, and S. J. Moorman, "Changes in gravitational force cause changes in gene expression in the lens of developing zebrafish," *Developmental Dynamics*, 235(10), 2686-2694 (2006).
- [10] L. Zhu, L. Nie, S. Xie *et al.*, "Attenuation of Antiviral Immune Response Caused by Perturbation of TRIM25-Mediated RIG-I Activation under Simulated Microgravity," *Cell Reports*, 34(1), 108600 (2021).
- [11] X. Hang, Y. Sun, M. Zhang *et al.*, "Effects of simulated-microgravity on zebrafish embryonic development and microRNA expression," 39th COSPAR Scientific Assembly, 39, 718 (2012).
- [12] C. Morabito, N. Steimberg, G. Mazzoleni *et al.*, "RCCS bioreactor-based modelled microgravity induces significant changes on in vitro 3D neuroglial cell cultures," *BioMed research international*, 2015, (2015).

- [13] S. J. Moorman, C. Burrell, R. Cordova *et al.*, "Stimulus dependence of the development of the zebrafish (*Danio rerio*) vestibular system," *Journal of neurobiology*, 38(2), 247-258 (1999).
- [14] T. Liu, X. Li, X. Sun *et al.*, "Analysis on forces and movement of cultivated particles in a rotating wall vessel bioreactor," *Biochemical engineering journal*, 18(2), 97-104 (2004).
- [15] H. C. Jo, R. J. Zawadzki, and D. Y. Kim, "In-vivo 3D zebrafish imaging for longitudinal observation of the retina regeneration with different light intensity irradiations." 11623, 1162325.